KONGU ENGINEERING COLLEGE

TRANSFORM YOURSELF

DEPARTMENT OF MECHANICAL ENGINEERING

MECHATRONICS & IOT
24MEC51

EXERCISE NO: 12

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MECHATRONICS & IoT

ASSIGNMENT REPORT – 12

TITLE:

Design and develop an Automated Color-Based Product Sorting System for Manufacturing Industries. The system automatically detects and sorts products based on color for industrial automation applications.

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Project Overview:

The Automated Color-Based Product Sorting System is designed to automatically identify and sort products according to their color. The system uses a TCS3200 color sensor to detect the color of an object and an ESP8266 NodeMCU as the main controller.

A servo motor is used as the sorting mechanism. When a product passes in front of the color sensor, the sensor detects its color and sends the information to the ESP8266. The controller processes the detected color and rotates the servo motor to direct the product into the appropriate sorting section.

The system demonstrates the application of industrial automation, sensor-based inspection, and automatic material handling.

Problem Statement

In many manufacturing industries, products need to be separated according to color before packaging, assembly, or further processing. Manual color-based sorting can be:

- Slow
- Labour-intensive
- Inconsistent
- Prone to human error
- Difficult to operate continuously
- Less suitable for high-speed production

Therefore, an automated color-based sorting system is required to identify products accurately and automatically separate them according to their color.

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Proposed Solution

The proposed system uses a TCS3200 color **sensor** to identify the color of products moving through the sorting area.

The ESP8266 NodeMCU receives the sensor output and determines the product color.

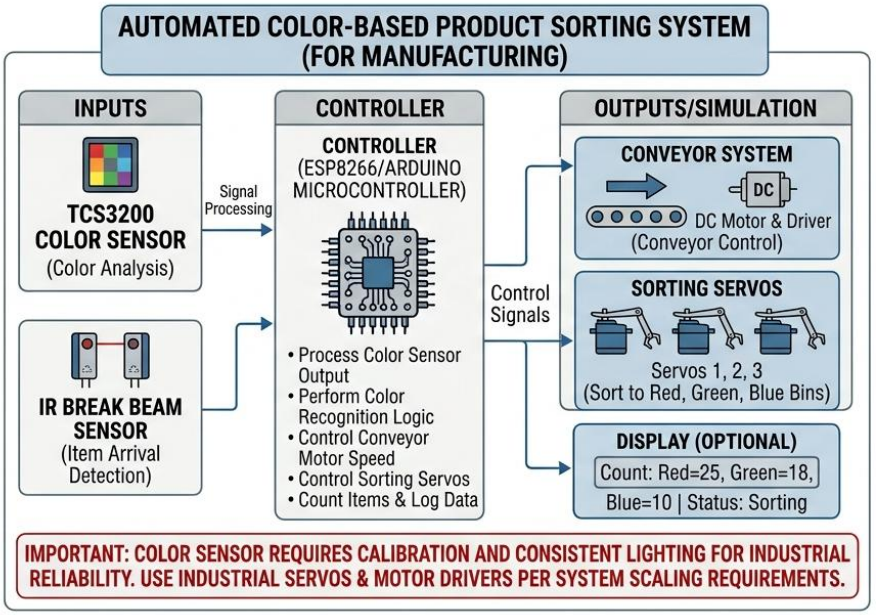
A servo motor acts as the sorting mechanism.

Example:

```
RED PRODUCT    → SERVO POSITION 1 → RED BIN
GREEN PRODUCT  → SERVO POSITION 2 → GREEN BIN
BLUE PRODUCT   → SERVO POSITION 3 → BLUE BIN
```

LED indicators can be used to show the detected color.

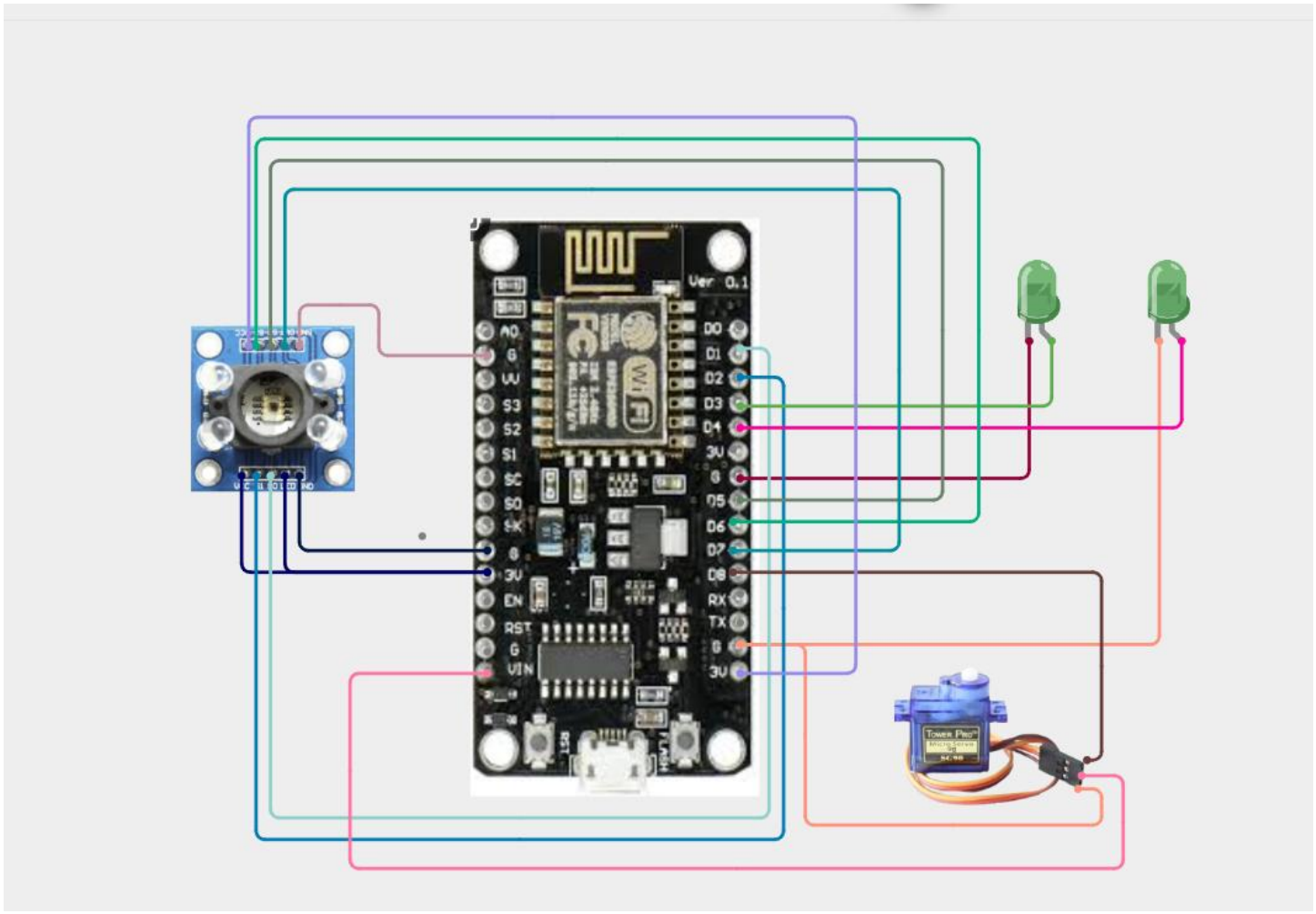
System Architecture



Circuit Table

S.No.	Component	Pin	ESP8266 Connection
1	TCS3200	VCC	Suitable supply
2	TCS3200	GND	GND
3	TCS3200	S0	D1
4	TCS3200	S1	D2
5	TCS3200	S2	D5
6	TCS3200	S3	D6
7	TCS3200	OUT	D7
8	Servo Motor	Signal	D4
9	Servo Motor	VCC	External 5 V supply
10	Servo Motor	GND	Common GND
11	Red LED	Anode	D0 through 220 Ω
12	Red LED	Cathode	GND
13	Green LED	Anode	D3 through 220 Ω
14	Green LED	Cathode	GND
15	Blue LED	Anode	Suitable GPIO through 220 Ω
16	Blue LED	Cathode	GND
17	Buzzer	Positive	Suitable GPIO
18	Buzzer	Negative	GND

Circuit Design



Algorithm

1. START

2. Initialize ESP8266 NodeMCU.

3. Initialize TCS3200 color sensor.

4. Initialize servo motor.

5. Initialize LED indicators and buzzer.

6. Set the initial servo position.

7. Wait for a product to be placed
   in front of the color sensor.

8. Read the RED component.

9. Read the GREEN component.

10. Read the BLUE component.

11. Compare the RGB values.

12. Identify the product color.

13. If RED is detected:
    Turn ON Red LED.
    Move servo to Red position.
    Direct product to Red bin.

14. Else if GREEN is detected:
    Turn ON Green LED.
    Move servo to Green position.
    Direct product to Green bin.

15. Else if BLUE is detected:
    Turn ON Blue LED.

Move servo to Blue position.
Direct product to Blue bin.

16. Else:
Turn ON buzzer.
Identify product as UNKNOWN.
Send product to reject/unknown bin.

17. Return servo to its initial position.

18. Wait for the next product.

19. Repeat the process continuously.

20. STOP.

Coding

```
#include <Servo.h>

#include <ESP8266WiFi.h>

#include "AdafruitIO_WiFi.h"



//================ WiFi =================

#define WIFI_SSID "Harish Kumar K K"

#define WIFI_PASS "Mavboiii@10"



//============== Adafruit IO ============

#define IO_USERNAME "Jr_Frost"

#define IO_KEY "aio_obVD87ub3BbEG4osjIOQjHNrUn7m"
```

```
AdafruitIO_WiFi io(IO_USERNAME, IO_KEY, WIFI_SSID, WIFI_PASS);

// Adafruit Feeds
AdafruitIO_Feed *redFeed = io.feed("red-value");
AdafruitIO_Feed *greenFeed = io.feed("green-value");
AdafruitIO_Feed *blueFeed = io.feed("blue-value");
AdafruitIO_Feed *colorFeed = io.feed("detected-color");

//============== TCS3200 ================
#define S0_PIN D1
#define S1_PIN D2
#define S2_PIN D3
#define S3_PIN D4
#define SENSOR_OUT D6

//================ Servo ================
#define SERVO_PIN D5

//================ LEDs ================
#define RED_LED D7
#define GREEN_LED D0
```

```
#define BLUE_LED D8

//============== Servo Angles ============
#define HOME_ANGLE 90
#define RED_ANGLE 30
#define GREEN_ANGLE 90
#define BLUE_ANGLE 150

Servo sortingServo;

//=========================================
// Setup
//=========================================

void setup() {

  Serial.begin(115200);

  pinMode(S0_PIN, OUTPUT);
  pinMode(S1_PIN, OUTPUT);
  pinMode(S2_PIN, OUTPUT);
```

```
pinMode(S3_PIN, OUTPUT);

pinMode(SENSOR_OUT, INPUT);

digitalWrite(S0_PIN, HIGH);
digitalWrite(S1_PIN, LOW);

pinMode(RED_LED, OUTPUT);
pinMode(GREEN_LED, OUTPUT);
pinMode(BLUE_LED, OUTPUT);

turnOffLEDs();

sortingServo.attach(SERVO_PIN);
sortingServo.write(HOME_ANGLE);

Serial.println("Connecting to Adafruit IO...");

io.connect();
```

```
  while (io.status() < AIO_CONNECTED) {
    Serial.print(".");
    delay(500);

  }

  Serial.println();
  Serial.println("Connected to Adafruit IO!");
  Serial.println(io.statusText());

  delay(1000);

  Serial.println("System Ready");
}

//============================================

void turnOffLEDs() {
  digitalWrite(RED_LED, LOW);
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(BLUE_LED, LOW);
```

```
}
//=========================================
unsigned long readSensor(bool s2, bool s3) {
 digitalWrite(S2_PIN, s2);

  digitalWrite(S3_PIN, s3);

  delay(30);

  return pulseIn(SENSOR_OUT, LOW, 100000);
}

unsigned long readRed() {
  return readSensor(LOW, LOW);
}

unsigned long readGreen() {
  return readSensor(HIGH, HIGH);
}

unsigned long readBlue() {
```

```
  return readSensor(LOW, HIGH);
}

//=========================================

void uploadValues(unsigned long r, unsigned long g, unsigned long b) {

  redFeed->save((int)r);

  greenFeed->save((int)g);
  blueFeed->save((int)b);
}

//=========================================

void sortRed() {

Serial.println("RED DETECTED");

  colorFeed->save("RED");
```

```
  turnOffLEDs();
  digitalWrite(RED_LED, HIGH);

  sortingServo.write(RED_ANGLE);
  delay(1500);
  sortingServo.write(HOME_ANGLE);

  delay(500);
  digitalWrite(RED_LED, LOW);

}

//=======================================

void sortGreen() {

  Serial.println("GREEN DETECTED");

  colorFeed->save("GREEN");
```

```
  turnOffLEDs();
  digitalWrite(GREEN_LED, HIGH);

  sortingServo.write(GREEN_ANGLE);
  delay(1500);

  sortingServo.write(HOME_ANGLE);

  delay(500);
  digitalWrite(GREEN_LED, LOW);


}

//=======================================

void sortBlue() {

  Serial.println("BLUE DETECTED");

  colorFeed->save("BLUE");
```

```
  turnOffLEDs();
  digitalWrite(BLUE_LED, HIGH);

  sortingServo.write(BLUE_ANGLE);
  delay(1500);

  sortingServo.write(HOME_ANGLE);

  delay(500);

  digitalWrite(BLUE_LED, LOW);
}

//=======================================

void unknownColor() {

  Serial.println("UNKNOWN COLOR");

  colorFeed->save("UNKNOWN");
```

```
  turnOffLEDs();

  sortingServo.write(HOME_ANGLE);

  delay(500);
}

// Main Loop

void loop() {
  io.run();   // Keep Adafruit connection alive

  unsigned long red;
  unsigned long green;
  unsigned long blue;

  Serial.println();
  Serial.println("Reading color...");

  red = readRed();
  delay(20);
```

```
green = readGreen();
delay(20);

blue = readBlue();

Serial.print("RED   = ");

Serial.println(red);



Serial.print("GREEN = ");
Serial.println(green);



Serial.print("BLUE  = ");
Serial.println(blue);

uploadValues(red, green, blue);

if (red == 0 || green == 0 || blue == 0) {

  Serial.println("Invalid reading");
```

```
  delay(500);
  return;
}

if ((red < green * 0.75) && (red < blue * 0.75)) {
  sortRed();

}

else if ((green < red * 0.80) && (green < blue * 0.80)) {

  sortGreen();
}

else if ((blue < red * 0.75) && (blue < green * 0.75)) {

  sortBlue();
}

else {
```

```
  unknownColor();
 }


 Serial.println("----------------------------");


 delay(1000);
}
```

System Working

Step 1 – Product Detection

A product is placed in front of the TCS3200 sensor.

The sensor detects the reflected light from the product.

Step 2 – Color Measurement

The TCS3200 measures the RGB components of the reflected light.

The ESP8266 reads the sensor's output frequency.

Step 3 – Color Identification

The controller compares the RGB values.

For example:

Red value > Green value

Red value > Blue value

↓

RED PRODUCT

Step 4 – Sorting Decision

After identifying the color, the ESP8266 sends a control signal to the servo motor.

Step 5 – Product Sorting

The servo moves the sorting arm to the appropriate position.

Red → Red Bin

Green → Green Bin

Blue → Blue Bin

Step 6 – Servo Reset

After sorting, the servo returns to its initial position and waits for the next product.

Bills Of Material

S.No.	Component	Specification	Quantity	Purpose
1	ESP8266 NodeMCU	Wi-Fi Microcontroller	1	Main Controller
2	TCS3200 Color Sensor	RGB Color Sensor	1	Color Detection
3	Servo Motor	SG90	1	Sorting Mechanism
4	Red LED	5 mm	1	Red Indication
5	Green LED	5 mm	1	Green Indication
6	Blue LED	5 mm	1	Blue Indication
7	Resistor	220 Ω	3	LED Protection
8	Buzzer	3.3/5 V	1	Alert
9	Breadboard	Standard	1	Prototype Assembly
10	Jumper Wires	Male-Male	1 Set	Connections
11	USB Cable	Micro USB	1	Programming/Power
12	External Power Supply	Suitable 5 V	1	Servo Power

Prototype Implementation

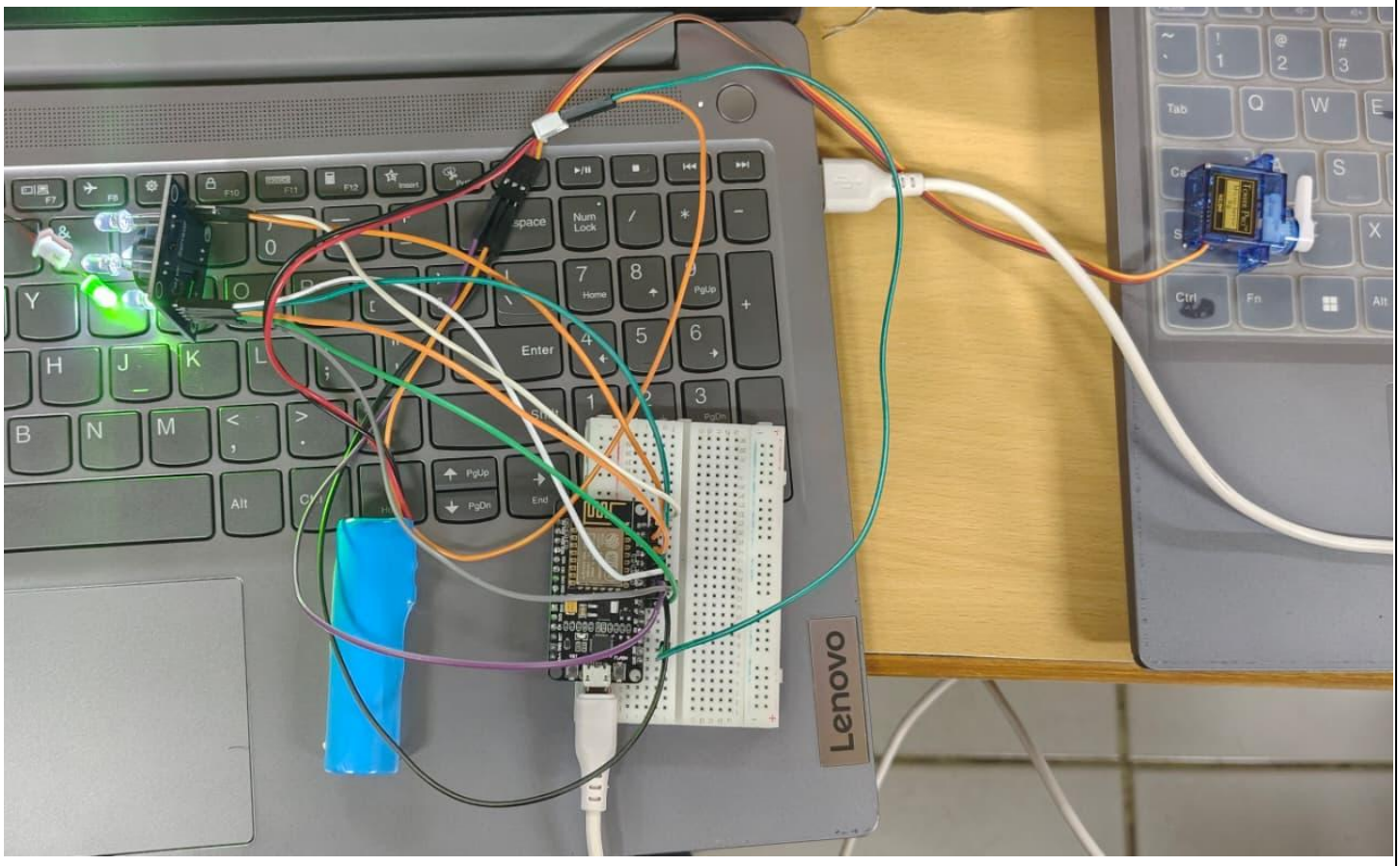
The prototype consists of an ESP8266 NodeMCU, TCS3200 color sensor, servo motor, LEDs, buzzer, breadboard, and jumper wires.

The TCS3200 is positioned above or beside the product path. When a product reaches the sensing position, the sensor detects its color.

The ESP8266 processes the sensor readings and identifies the product.
The servo motor then moves the sorting mechanism to the corresponding position.

After the product is sorted, the servo returns to its neutral position and the system waits for the next product.

The prototype can be extended by adding a DC motor and conveyor belt to create a continuous industrial sorting line.



Results & Performance Analysis

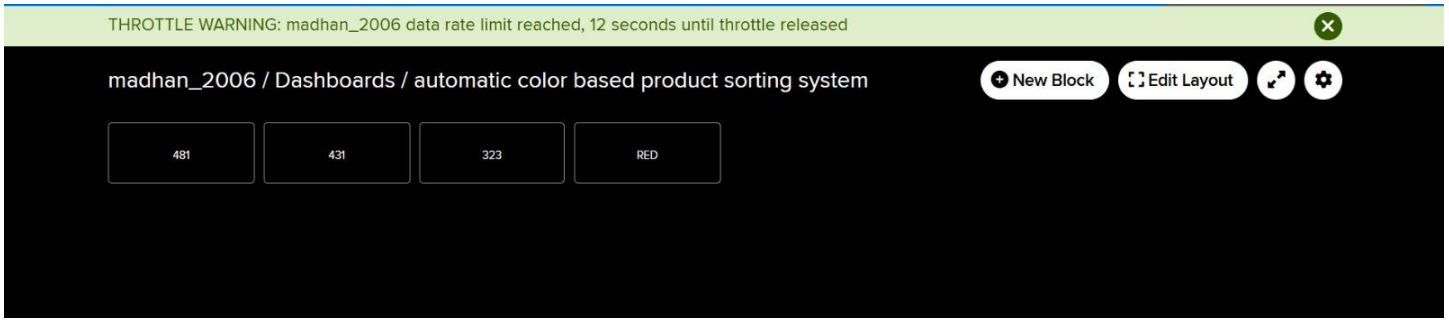
The prototype successfully demonstrates automatic color identification and product sorting.

Parameter	Result
Product Color Detection	Yes
Red Detection	Yes
Green Detection	Yes
Blue Detection	Yes
Automatic Sorting	Yes
Servo Control	Yes
Visual Indication	Yes
Unknown Color Detection	Supported
Continuous Operation	Supported

Performance Analysis

The system reduces the need for manual product sorting and provides consistent sorting based on the programmed color thresholds.
Performance can be improved by:

- Calibrating the TCS3200 sensor.
- Maintaining constant illumination.
- Keeping the sensor at a fixed distance.
- Using a controlled conveyor speed.
- Filtering noisy sensor readings.
- Using multiple readings before making a sorting decision.




```
BLUE  = 468
RED DETECTED
------------------------------

Reading color...
RED   = 300
GREEN = 411
BLUE  = 447
RED DETECTED
------------------------------

Reading color...
RED   = 290
GREEN = 313
BLUE  = 240
UNKNOWN COLOR
------------------------------

Reading color...
RED   = 295
GREEN = 262
BLUE  = 155
BLUE DETECTED
------------------------------

Reading color...
RED   = 279
GREEN = 258
BLUE  = 158
BLUE DETECTED
------------------------------

Reading color...
RED   = 277
GREEN = 268
BLUE  = 203
UNKNOWN COLOR
------------------------------

Reading color...
RED   = 282
GREEN = 282
BLUE  = 189
BLUE DETECTED
------------------------------

Reading color...
RED   = 295
GREEN = 253
BLUE  = 152
BLUE DETECTED
------------------------------
```

```
Reading color...
RED   = 93
GREEN = 49
BLUE  = 106
GREEN DETECTED
------------------------------

Reading color...
RED   = 313
GREEN = 425
BLUE  = 468
RED DETECTED
```

```
RED   = 462
GREEN = 562
BLUE  = 607
UNKNOWN COLOR
------------------------------

Reading color...
RED   = 296
GREEN = 408
BLUE  = 474
RED DETECTED
------------------------------

Reading color...
RED   = 319
GREEN = 447
BLUE  = 500
RED DETECTED
------------------------------

Reading color...
RED   = 312
GREEN = 408
BLUE  = 465
UNKNOWN COLOR
------------------------------

Reading color...
RED   = 182
GREEN = 120
BLUE  = 197
GREEN DETECTED
------------------------------

Reading color...
RED   = 120
GREEN = 72
BLUE  = 145
GREEN DETECTED
------------------------------

Reading color...
RED   = 96
GREEN = 49
BLUE  = 111
GREEN DETECTED
------------------------------

Reading color...
RED   = 85
GREEN = 45
BLUE  = 102
GREEN DETECTED
```